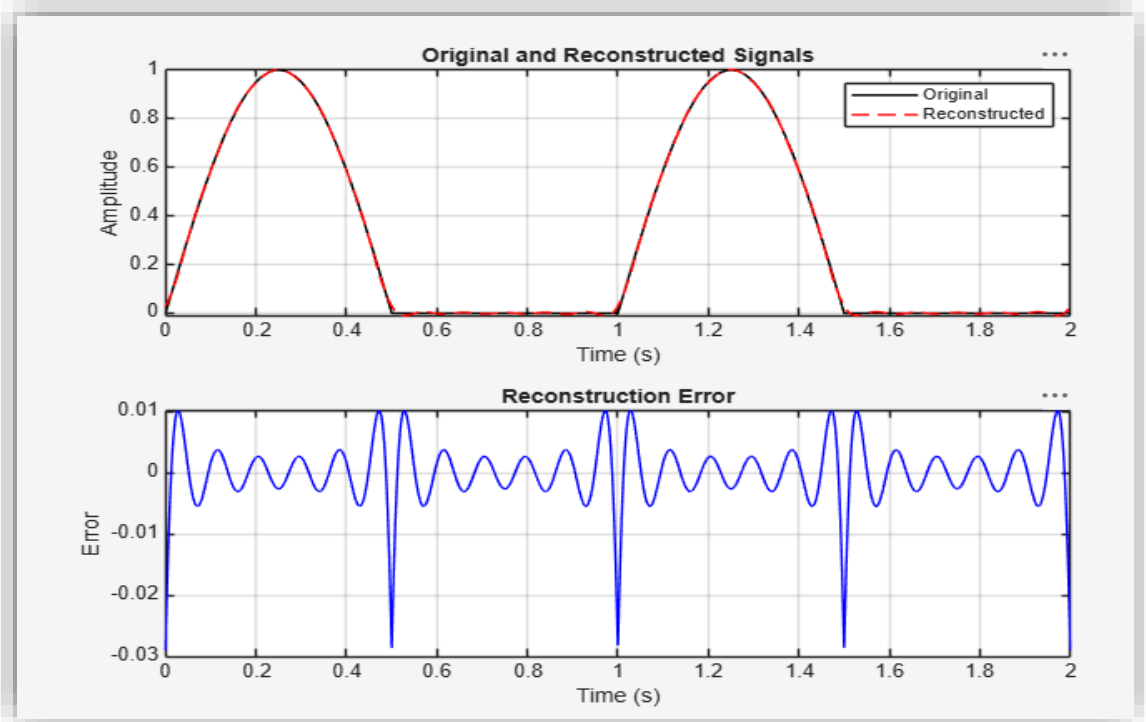
Experiment 5
Fourier Series Analysis of Periodic Signals

(iv)

```
clc;
clear;
close all;
figure;
f0 = 1;
T = 1/f0;
t = linspace(0,2*T,4000);
x = max(sin(2*pi*f0*t),0);
N = 10;
a0 = (2/T)*trapz(t(1:2000),x(1:2000));
an = zeros(1,N); bn = zeros(1,N);
tt = t(1:2000); xx = x(1:2000);
for k = 1:N
    an(k) = (2/T)*trapz(tt,xx.*cos(2*pi*k*f0*tt));
    bn(k) = (2/T)*trapz(tt,xx.*sin(2*pi*k*f0*tt));
end
xr = a0/2*ones(size(t));
for k = 1:N
    xr = xr + an(k)*cos(2*pi*k*f0*t) + bn(k)*sin(2*pi*k*f0*t);
end
err = x-xr;
subplot(2,1,1);
plot(t,x,'k',t,xr,'r--','LineWidth',1.1);
grid on; xlabel('Time (s)'); ylabel('Amplitude');
title('Original and Reconstructed Signals');
legend('Original','Reconstructed');
subplot(2,1,2);
plot(t,err,'b'); grid on;
xlabel('Time (s)'); ylabel('Error');
title('Reconstruction Error');
```

Code



Output